



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians with username and password"
- "Validate password strength and reject weak passwords"
- "Prevent duplicate librarian registration"
- "Authenticate librarian credentials securely"
- "Validate librarian sessions and handle access denial"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books by title and prevent duplicate entries"
- "Delete registered books"
- "Consult books by identifier or title"
- "List all registered books"
- "List books available for loan"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans and prevent duplicate entries"
- "Delete registered readers"
- "Consult readers by identifier or name"
- "List all registered readers"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans with loan date for registered readers"
- "Validate reader eligibility and book availability during loan creation"
- "Ensure each book is loaned to only one reader at a time"
- "Permit multiple loans for the same reader"
- "Remove loan links when associated books or readers are deleted"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear system data and base structure"
- "Recreate base structure to restore a clean environment state"
- "Coordinate functional areas to ensure consistent data reset for tests"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two user-provided integers"
- "Calculate Fibonacci sequence values using multiple approaches"
- "Calculate Ackermann function results using multiple approaches"
- "Provide a service catalog to consult and test available endpoints"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record relevant operations from registration"

dependencies:

- "authentication"

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"

to: "Audit Service"

- from: "Book Catalog Service"
to: "Loan Management Service"

- from: "Book Catalog Service"
to: "Audit Service"

- from: "Reader Management Service"
to: "Loan Management Service"

- from: "Reader Management Service"
to: "Audit Service"

- from: "Loan Management Service"
to: "Book Catalog Service"

- from: "Loan Management Service"
to: "Reader Management Service"

- from: "Loan Management Service"
to: "Audit Service"

- from: "Environment Management Service"
to: "Authentication Service"

- from: "Environment Management Service"
to: "Book Catalog Service"

- from: "Environment Management Service"
to: "Reader Management Service"

- from: "Environment Management Service"
to: "Loan Management Service"

- from: "Audit Service"
to: "authentication"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Create customer usage sessions"
- "Validate active sessions"
- "Invalidate sessions upon logout"
- "Manage session expiration"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Manage registered airports and routes"
- "Search one-way and round-trip flights"

- "Present flight schedules and reference fares"

dependencies:

- ""

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create one-way and round-trip flight bookings"
- "Generate booking identifiers and record booking dates"
- "Consult and list customer bookings"
- "Cancel active bookings"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Retrieve customer profile data"
- "Update customer contact and address details"
- "Enforce password confirmation for profile modifications"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Consult customer loyalty program status"
- "Retrieve annual and total accumulated miles"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload database with sample operational data"
- "Provide summarized operational metrics and indicators"
- "Coordinate operational data reset"

dependencies:

- "Authentication Service"
- "Flight Search Service"
- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Flight Search Service"
- from: "Environment Administration Service"
to: "Booking Service"
- from: "Environment Administration Service"
to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Location Management Service"

responsibilities:

- "Register and maintain logistics locations"
- "Provide controlled list of registered locations"
- "Validate location existence and provide geographic coordinates"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Voyage Management Service"
- "Logistics Event Registration Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/location-management-service"

- name: "Voyage Management Service"

responsibilities:

- "Register and maintain voyages and carrier movements"
- "Maintain controlled list of voyages"
- "Provide voyage legs with load points and unload points"
- "Validate voyage existence for routing and event registration"

dependencies:

- "Location Management Service"
- "Route Planning Service"
- "Logistics Event Registration Service"

suggested_endpoints:

- "/api/voyage-management-service"

- name: "Cargo Management Service"

responsibilities:

- "Register new cargo with origin destination and arrival deadline"
- "Assign unique tracking identifiers to cargo"
- "Manage cargo lifecycle and routing status"
- "Update cargo destination and trigger route reevaluation"
- "Validate origin and destination against registered locations"

dependencies:

- "Location Management Service"
- "Route Planning Service"
- "Cargo Tracking Service"
- "Logistics Event Registration Service"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieve and generate candidate routes meeting origin destination and deadline"
- "Validate route options and schedule compatibility"
- "Display route legs with load points unload points voyages and expected dates"
- "Replan routes for cargos off route or with changed destinations"

dependencies:

- "Cargo Management Service"
- "Voyage Management Service"
- "Location Management Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Registration Service"

responsibilities:

- "Register logistics events including reception load unload customs and pickup"
- "Validate event association with cargo location and voyage"
- "Record event occurrence date and registration timestamp"
- "Isolate and record rejected event registration attempts for analysis"
- "Notify valid event occurrences to tracking"

dependencies:

- "Cargo Management Service"
- "Location Management Service"
- "Voyage Management Service"

- "Cargo Tracking Service"

- "Integration Service"

suggested_endpoints:

- "/api/logistics-event-registration-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consolidate cargo operational status transport status and event history"

- "Recalculate last known location next expected step and estimated arrival"

- "Detect and signal off-route operational deviations"

- "Provide cargo tracking views with geographic support highlighting origin destination and last known loc

- "Distinguish delivered in-transit and deviated cargos"

dependencies:

- "Cargo Management Service"

- "Route Planning Service"

- "Logistics Event Registration Service"

- "Location Management Service"

- "Integration Service"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Ingest logistics events from operational interface integration services and file processing"

- "Process input files periodically without manual intervention"

- "Separate invalid input records for handling"

- "Provide consolidated cargo tracking views to external monitoring channels"

- "Exchange cargo information in standardized formats"

dependencies:

- "Logistics Event Registration Service"

- "Cargo Tracking Service"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Location Management Service"

- to: "Cargo Management Service"

- from: "Location Management Service"

- to: "Route Planning Service"

- from: "Location Management Service"

- to: "Voyage Management Service"

- from: "Location Management Service"

- to: "Logistics Event Registration Service"

- from: "Location Management Service"

- to: "Cargo Tracking Service"

- from: "Voyage Management Service"

to: "Location Management Service"

- from: "Voyage Management Service"

to: "Route Planning Service"

- from: "Voyage Management Service"

to: "Logistics Event Registration Service"

- from: "Cargo Management Service"

to: "Location Management Service"

- from: "Cargo Management Service"

to: "Route Planning Service"

- from: "Cargo Management Service"

to: "Cargo Tracking Service"

- from: "Cargo Management Service"

to: "Logistics Event Registration Service"

- from: "Route Planning Service"

to: "Cargo Management Service"

- from: "Route Planning Service"

to: "Voyage Management Service"

- from: "Route Planning Service"

to: "Location Management Service"

- from: "Route Planning Service"

to: "Cargo Tracking Service"

- from: "Logistics Event Registration Service"

to: "Cargo Management Service"

- from: "Logistics Event Registration Service"

to: "Location Management Service"

- from: "Logistics Event Registration Service"

to: "Voyage Management Service"

- from: "Logistics Event Registration Service"

to: "Cargo Tracking Service"

- from: "Logistics Event Registration Service"

to: "Integration Service"

- from: "Cargo Tracking Service"

to: "Cargo Management Service"

- from: "Cargo Tracking Service"

to: "Route Planning Service"

- from: "Cargo Tracking Service"

to: "Logistics Event Registration Service"

- from: "Cargo Tracking Service"

to: "Location Management Service"

- from: "Cargo Tracking Service"

to: "Integration Service"

- from: "Integration Service"

to: "Logistics Event Registration Service"

- from: "Integration Service"
to: "Cargo Tracking Service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials"
- "Manage investor login and active sessions"
- "End investor sessions upon logout"
- "Validate authentication for restricted resource access"

dependencies:

- "Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Account Service"

responsibilities:

- "Register new investors"
- "Validate investor registration and profile updates"
- "Maintain investor account and profile details"
- "Manage investor cash balance"
- "Record access dates and login statistics"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/account-service"

- name: "Market Data Service"

responsibilities:

- "Query single and multiple asset quotes"
- "Provide consolidated market summary"
- "Update traded volume and market indicators following operations"
- "Handle invalid asset codes"

dependencies:

- ""

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create buy and sell trading orders"
- "Calculate and apply fixed operation fees"
- "Manage trading order lifecycle and status tracking"
- "Process orders immediately or deferred according to simulation configuration"

- "Maintain and consult investor order history"
- "Alert investors about completed orders"

dependencies:

- "Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor portfolio positions"
- "Register positions upon completed buy orders"
- "Remove positions upon completed sell orders"
- "Calculate invested amount and market value and gains or losses"
- "Provide consolidated financial position summary"
- "Initiate sales directly from portfolio positions"

dependencies:

- "Account Service"
- "Market Data Service"
- "Trading Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Prepare and rebuild simulation database"
- "Reset simulation operational data"
- "Provide summarized execution statistics and monitoring indicators"

dependencies:

- "Account Service"
- "Market Data Service"
- "Trading Service"
- "Portfolio Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Account Service"
- from: "Account Service"
to: "Authentication Service"
- from: "Trading Service"
to: "Account Service"

- from: "Trading Service"
to: "Market Data Service"

- from: "Trading Service"
to: "Portfolio Service"

- from: "Trading Service"
to: "Simulation Administration Service"

- from: "Portfolio Service"
to: "Account Service"

- from: "Portfolio Service"
to: "Market Data Service"

- from: "Portfolio Service"
to: "Trading Service"

- from: "Simulation Administration Service"
to: "Account Service"

- from: "Simulation Administration Service"
to: "Market Data Service"

- from: "Simulation Administration Service"
to: "Trading Service"

- from: "Simulation Administration Service"
to: "Portfolio Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users"
- "Authenticate user credentials"
- "Manage user sessions"
- "Control user profiles and permissions"
- "Validate access rights and deny unauthorized requests"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "Register new movies"
- "Delete movies individually and in batch"
- "List movies with pagination"
- "Provide movie details and full cast"
- "Manage and list movie categories"
- "Filter movies by category"

- "Enforce movie uniqueness"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload video files"
- "Store and manage video files in local storage"
- "Deliver video files for playback"
- "Provide movie download links"
- "Verify video file availability for consumption"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Organize products by animal categories"
- "Display product lists and available items"
- "Provide item details including description and price"
- "Search products by keyword"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register customer identification"

dependencies:

- "contact"

suggested_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Manage customer login and session termination"
- "Control and validate authorized access to protected features"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Manage temporary selection of items in the cart"
- "Consolidate items and update or remove item quantities"
- "Calculate cart subtotal automatically"
- "Clear cart items after order completion"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Track item stock availability and backorder status"
- "Consult stock status during the purchase journey"
- "Register stock decrease upon order completion"

dependencies:

- "Catalog Service"

suggested_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Conduct checkout process and finalize purchases"
- "Compose order using cart items and customer registration data"
- "Manage billing"

dependencies:

- "delivery addresses"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"

to: "Inventory Service"

- from: "Customer Account Service"
to: "contact"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"
to: "Catalog Service"
- from: "Order Service"
to: "delivery addresses"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owners with contact information"
- "Update existing owner details"
- "Search owners by last name with pagination"
- "Retrieve detailed owner profiles"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"
- "Update registered animal details"
- "Provide predefined list of animal types"
- "Validate animal name uniqueness per owner and birth date"
- "Maintain animal records and owner associations"

dependencies:

- "Owner Service"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register visits for specific animals"
- "Maintain animal visit and consultation history"
- "Coordinate visit registration with veterinarian assignment"
- "Ensure visit persistence and internal process resumption"

dependencies:

- "Animal Service"

- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "List veterinarians and specialties with pagination"
- "Calculate veterinarian workload and next expected visits"
- "Automatically assign visits prioritizing lowest future workload"
- "Provide structured veterinarian data for external systems"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"
to: "Owner Service"
- from: "Visit Service"
to: "Animal Service"
- from: "Visit Service"
to: "Veterinarian Service"
- from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records"
- "Manage individual user configurations"
- "Provide basic configuration parameters"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "CRM and Proposals Service"
- "Billing Service"
- "Notification Service"

suggested_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain user profiles and work agreements"
- "Manage password changes and resets"

- "Control access by user profile and scope"
- "Provide user validation"

dependencies:

- "Administration and Configuration Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "Content and Document Management Service"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements and vacation requests"
- "Control vacation request states and calculate balances"
- "Monitor contract expirations and vacation notifications"

dependencies:

- "Identity and Access Service"
- "Administration and Configuration Service"
- "Notification Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project and Work Tracking Service"

responsibilities:

- "Maintain projects linked to clients"
- "Register internal and external activities"
- "Classify activities and calculate worked hours"
- "Enforce evidence policies"

dependencies:

- "Workforce Management Service"
- "Identity and Access Service"
- "File Storage Service"
- "Notification Service"

suggested_endpoints:

- "/api/project-and-work-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations and contacts"
- "Manage commercial interactions and proposals"
- "Control proposal states and composition"
- "Link proposals to projects"

dependencies:

- "Administration and Configuration Service"
- "Project and Work Tracking Service"

suggested_endpoints:

- "/api/crm-and-proposals-service"
- name: "Financial Management Service"
- responsibilities:
 - "Maintain financial accounts and entries"
 - "Track financial indicators"
 - "Import billable hours and costs"
- dependencies:
 - "Billing Service"
 - "Project and Work Tracking Service"
- suggested_endpoints:
 - "/api/financial-management-service"
- name: "Billing Service"
- responsibilities:
 - "Detail invoices and calculate totals"
- dependencies:
 - "taxes"
- suggested_endpoints:
 - "/api/billing-service"
- name: "Content and Document Management Service"
- responsibilities:
 - "Organize documents by categories and versions"
 - "Associate documents with users and areas"
- dependencies:
 - "File Storage Service"
 - "Identity and Access Service"
- suggested_endpoints:
 - "/api/content-and-document-management-service"
- name: "Commissioning Service"
- responsibilities:
 - "Register and manage commissioning orders"
 - "Process commission workflows"
- dependencies:
 - "Project and Work Tracking Service"
 - "Workforce Management Service"
 - "File Storage Service"
 - "Notification Service"
- suggested_endpoints:
 - "/api/commissioning-service"
- name: "Reporting Service"
- responsibilities:
 - "Consolidate cross-domain metrics"
 - "Export analytical reports"
- dependencies:

- "Workforce Management Service"
- "Project and Work Tracking Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"
- "Manage notification settings"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store files"
- "Retrieve files"
- "Manage storage configurations"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Administration and Configuration Service"
to: "Identity and Access Service"
- from: "Administration and Configuration Service"
to: "Workforce Management Service"
- from: "Administration and Configuration Service"
to: "CRM and Proposals Service"
- from: "Administration and Configuration Service"
to: "Billing Service"
- from: "Administration and Configuration Service"
to: "Notification Service"
- from: "Identity and Access Service"
to: "Administration and Configuration Service"
- from: "Identity and Access Service"
to: "Workforce Management Service"
- from: "Identity and Access Service"
to: "Project and Work Tracking Service"
- from: "Identity and Access Service"

to: "Content and Document Management Service"

- from: "Workforce Management Service"

to: "Identity and Access Service"

- from: "Workforce Management Service"

to: "Administration and Configuration Service"

- from: "Workforce Management Service"

to: "Notification Service"

- from: "Project and Work Tracking Service"

to: "Workforce Management Service"

- from: "Project and Work Tracking Service"

to: "Identity and Access Service"

- from: "Project and Work Tracking Service"

to: "File Storage Service"

- from: "Project and Work Tracking Service"

to: "Notification Service"

- from: "CRM and Proposals Service"

to: "Administration and Configuration Service"

- from: "CRM and Proposals Service"

to: "Project and Work Tracking Service"

- from: "Financial Management Service"

to: "Billing Service"

- from: "Financial Management Service"

to: "Project and Work Tracking Service"

- from: "Billing Service"

to: "taxes"

- from: "Content and Document Management Service"

to: "File Storage Service"

- from: "Content and Document Management Service"

to: "Identity and Access Service"

- from: "Commissioning Service"

to: "Project and Work Tracking Service"

- from: "Commissioning Service"

to: "Workforce Management Service"

- from: "Commissioning Service"

to: "File Storage Service"

- from: "Commissioning Service"

to: "Notification Service"

- from: "Reporting Service"

to: "Workforce Management Service"

- from: "Reporting Service"

to: "Project and Work Tracking Service"

- from: "Reporting Service"

to: "CRM and Proposals Service"

- from: "Reporting Service"
to: "Financial Management Service"

- from: "Reporting Service"
to: "Billing Service"

- from: "Reporting Service"
to: "Commissioning Service"

- from: "Notification Service"
to: "Administration and Configuration Service"

- from: "File Storage Service"
to: "Administration and Configuration Service"